meter of judging the effectiveness of an artificial intelligence agent because it manages to test several parameters within a simple methodology.

In their book Artificial Intelligence: A Modern Approach, Stuart Russell and Peter Norvig write about the four base criteria that the Turing Test puts any AI through:

- **Natural language processing:** The most essential part of any artificial intelligence system would have to be its ability to converse with humans in a way that doesn't make it obvious that the other party is a machine. For this, the AI needs to both understand and reciprocate natural language. After all, what use would it be to have a robot who doesn't understand half of the things you're saying? Natural language processing is perhaps the most difficult part of building an AI agent, since it would have to take into account various dialects and slang, along with having to correlate different types of data to come up with the meaning of what was said.
- **Knowledge representation:** During a conversation, there's often a give-and-take of new information among humans. A successful AI has to have the ability to store and retrieve data it has just received from a new source, because that's key to carrying on a conversation. While it sounds difficult, this is an aspect that has advanced enough in most mechanical systems now.
- **Automated reasoning:** Reasoning, or the ability to draw rational conclusions from different data, is one of those problems that seems simple on the surface but is extremely complex the more you get into it. To store data on a system isn't a problem at all. But to give the system the ability to figure out which data correlates with which other data? That's where things start becoming complicated. And when you add the new information learned from knowledge representation to it, it's suddenly a Herculean task to come up with a system that can use old data to come up with new solutions.
- **Machine learning:** A related part of automated reasoning, machine learning is the next step in what would simply be called "pattern recognition". As humans, we have an innate ability to look at new things and make them fit into our understanding of the world, based on the patterns of information we have experienced in our lives. It's something we do without even thinking about it. But for a machine, there isn't an intuitive ability to apply patterns and detect new environments – so to pass the Turing Test would be a major triumph in that department.